

Probability Solutions - Continuous Random Variables

Question 3.9

The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X with density function

$$f(x) = 2(x + 2)/5, \quad 0 < x < 1, \quad \text{elsewhere } f(x) = 0$$

(a) Show that $P(0 < X < 1) = 1$

$$\begin{aligned} P(0 < X < 1) &= \int_0^1 f(x) dx = \int_0^1 2(x + 2) dx \\ &= \int_0^1 2(x + 2) dx = 2 \int_0^1 x dx + 2 \int_0^1 2 dx \\ &= 2 \left[\frac{x^2}{2} \right]_0^1 + 4 [x]_0^1 \\ &= (1^2 - 0^2) + 4(1 - 0) = 1 + 4 = 5 \end{aligned}$$

Note: If the PDF must integrate to 1, the given function should be normalized. Suppose the correct PDF is

$$f(x) = \frac{2(x + 2)}{5}, \quad 0 < x < 1$$

Then

$$P(0 < X < 1) = \int_0^1 \frac{2(x + 2)}{5} dx = 1$$

(b) Probability that more than 1/4 but fewer than 1/2 respond

$$\begin{aligned} P\left(\frac{1}{4} < X < \frac{1}{2}\right) &= \int_{1/4}^{1/2} f(x) dx = \int_{1/4}^{1/2} \frac{2(x + 2)}{5} dx \\ &= \frac{2}{5} \int_{1/4}^{1/2} (x + 2) dx = \frac{2}{5} \left[\frac{x^2}{2} + 2x \right]_{1/4}^{1/2} \\ &= \frac{2}{5} \left[\left(\frac{1}{8} + 1 \right) - \left(\frac{1}{32} + \frac{1}{2} \right) \right] = \frac{2}{5} [1.125 - 0.53125] = \frac{2}{5} \cdot 0.59375 \approx 0.2375 \end{aligned}$$

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Question 3.14

The waiting time between successive speeders is a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-8x}, & x \geq 0 \end{cases}$$

(a) Probability of waiting less than 12 minutes using CDF

Convert 12 minutes to hours: $12/60 = 0.2$ hours. Then

$$P(X < 0.2) = F(0.2) = 1 - e^{-8 \cdot 0.2} = 1 - e^{-1.6}$$

$$\approx 1 - 0.2019 = 0.7981$$

(b) Probability using PDF

The PDF is obtained by differentiating the CDF:

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} 0, & x < 0 \\ 8e^{-8x}, & x \geq 0 \end{cases}$$

$$\begin{aligned} P(X < 0.2) &= \int_0^{0.2} f(x) dx = \int_0^{0.2} 8e^{-8x} dx \\ &= [-e^{-8x}]_0^{0.2} = -e^{-1.6} + 1 = 1 - e^{-1.6} \approx 0.7981 \end{aligned}$$